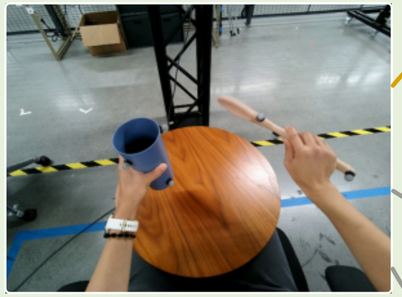
Egocentric Episode



colour camera +
mono stereo pair

Task & Objects via VLM

Qwen3-VL (local)

"The person holds the blue cup and stirs it with a wooden spoon, then puts both down."

"wooden spoon", "blue cup"

Hand Estimator

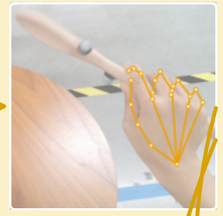
Dyn-HaMR

metric MANO trajectory

WiLoR stereo fallback (<2%)

hands out of objects

$\mathcal{M}_t^h$



FoundationStereo



table plane

metric geometry

DROID-SLAM

camera T_t, gravity $\mathbf{g}$

SAM 3 (text-prompted)



FoundationPose all ~252 hyp.

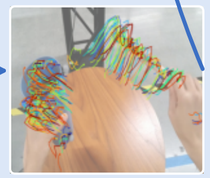
$\{X_t^k\}$



SAM 3D Objects + stereo scale



TrackEverything 3D tracks



$p_n(t)$

own meshes only - no provided scans, no mocap

Object Estimator

SelectTrack: interaction-aware refinement

Whole-sequence selection

$$\min_{k_{1:T}} \sum_t u_t(\mathcal{S}_t^o, X_t^{k_t}, p_n) + \lambda\, m_t(p_n)$$

Speed-limit flip filter

keep / flip / skip, ≤20° & 6 cm / frame

State-aware guards

$s_t \in \{\text{resting}, \text{held}\}$

rest hold · spin guard · twist unwrap

Contact sanity

objects rest on the table
hands pushed out of objects

4D HOI reconstruction



$\{\mathcal{M}_t^h,\ X_t^o,\ \mathcal{M}^o\}$

Kaggle rollouts + meshes



official evaluator → per-episode best run

FlashSAC per episode
task + contact reward
friction 1.0 / 1.5, 2 seeds

Retarget to Sharpa hands



Simulator grounding
settled starts · carved collision · cup bases

Robot Learning